Business Management HL · Set B · Paper 2

Worked mark scheme · 105 minutes · 50 marks

Award equivalent valid methods and conclusions. Apply the indicated mark allocation to each part. Credit consistent follow-through where the subsequent reasoning remains valid.

Section A · Bracken Sensors [12]

(a) **[4]**

Calculate current capacity utilization and fixed manufacturing overhead per unit if fixed overhead is allocated over actual output.

1 mark · Divide current output by normal capacity.

1 mark · 80%.

1 mark · Allocate CU 360,000 over 40,000 units.

1 mark · CU 9 per unit.

(b) **[3]**

Calculate the change in operating profit if Bracken accepts the order.

1 mark · Incremental contribution before freight is 8,000 × (35 − 28).

1 mark · CU 56,000.

1 mark · Net gain is CU 40,000 after shipping.

(c) **[3]**

Explain why an absorption-costing appraisal and a contribution appraisal could lead to different decisions.

1 mark · The allocated production cost of CU 37 per unit exceeds the offered price of CU 35.

1 mark · At spare capacity, allocated fixed costs remain payable with either decision.

1 mark · Contribution focuses on the extra revenue and avoidable costs of this order.

(d) **[2]**

Explain one reason spare capacity measured annually might overstate Bracken's ability to fulfil this order.

1 mark · Normal sales and the order might peak in the same weeks.

1 mark · Overtime, outsourcing or late delivery could reduce the calculated gain.

Section A · Mora Studio [10]

(a) **[4]**

Calculate current gearing and return on capital employed. Use non-current liabilities as a percentage of capital employed for gearing.

1 mark · Capital employed is CU 600,000.

1 mark · Gearing is 40%.

1 mark · Divide CU 84,000 by CU 600,000 and multiply by 100.

1 mark · ROCE is 14%.

(b) **[3]**

Calculate gearing after expansion under each financing option.

1 mark · Capital employed is CU 750,000 in either case.

1 mark · Debt option 390,000/750,000 = 52%.

1 mark · Equity option 240,000/750,000 = 32%.

(c) **[3]**

Calculate additional annual profit before tax after interest if Mora borrows. Explain one implication for risk.

1 mark · Interest is CU 12,000.

1 mark · Additional profit before tax is CU 3,000.

1 mark · A fall of more than CU 3,000 in project operating profit would eliminate the forecast gain although interest remains due.

Section A · Willow Joinery [8]

(a) **[3]**

Identify the critical path and calculate the minimum project duration.

1 mark · A–C ends after 9 days and B–D after 8.

1 mark · The critical path is A–C–E.

1 mark · Minimum duration is 12 working days.

(b) **[3]**

Determine total float on B and the effect of a two-day delay to B on overall completion.

1 mark · The B–D branch has 1 day of float, so B has 1 day total float.

1 mark · With the delay, B–D finishes at day 10 and E then takes 3.

1 mark · Completion moves to day 13, a one-day project delay.

(c) **[2]**

Explain one reason the critical path could change during the refit.

1 mark · An uncertain activity duration could lengthen the currently shorter B–D branch beyond 9 days.

1 mark · E would then wait for D, making that branch determine the completion date.

Section B · Option 1 · Senna Lamps [20]

(a) **[2]**

Calculate current inventory turnover in days.

1 mark · 36000/288000 multiplied by 360.

1 mark · 45 days.

(b) **[4]**

Calculate current creditor days and inventory days under the proposal.

1 mark · 24000/240000 multiplied by 360.

1 mark · Creditor days are 36.

1 mark · Divide the new inventory balance by the unchanged cost of sales and multiply by 360.

1 mark · New inventory days are 30.

(c) **[4]**

Find the change in the operating profit margin and explain the difference between the cash released from inventory and this change in profit.

1 mark · Revenue is CU 540,000 and margin falls from 10% to about 9.11%.

1 mark · A decrease of about 0.89 percentage points.

1 mark · CU 12,000 less average stock is a one-off release of working capital.

1 mark · The delivery charge reduces annual operating profit by CU 4,800.

(d) **[10]**

Evaluate whether Senna should adopt the new supply arrangement.

10 marks · Fifteen fewer inventory days release cash and free storage, but the recurring CU 4,800 cost exceeds the one-off CU 12,000 release after two and a half years in a simple undiscounted comparison if no other savings arise. That comparison mixes a balance-sheet release with a recurring expense, so assess the opportunity cost of cash and space, service reliability and stockout risk. A second source or minimum buffer may make the switch feasible. A cash-constrained firm can reasonably place a higher value on the near-term release. Use best fit. Award 0 for no creditable response, 1–2 for limited assertions, 3–4 for a partially explained point with weak application, 5–6 for developed analysis using case evidence, 7–8 for supported consideration of competing arguments and a reasoned decision, and 9–10 for sustained evaluation integrating specific evidence, a material limitation and a convincing judgement. Credit a well-supported alternative.

Section B · Option 2 · Fen Rail Events [20]

(a) **[2]**

Calculate the NPV of the venue in the busy scenario.

1 mark · 80,000 × 3.31 − 180,000.

1 mark · CU 84,800.

(b) **[4]**

Calculate the probability-weighted expected NPV for each option.

1 mark · Venue expected annual inflow is CU 62,000.

1 mark · Venue expected NPV is CU 25,220.

1 mark · Stage expected annual inflow is CU 39,800.

1 mark · Stage expected NPV is CU 31,738.

(c) **[4]**

Calculate both quiet-scenario NPVs and explain how these outcomes inform a risk-averse owner.

1 mark · Venue NPV is 35,000 × 3.31 − 180,000 = −CU 64,150.

1 mark · Stage NPV is 32,000 × 3.31 − 100,000 = CU 5,920.

1 mark · The stage stays positive in the modelled quiet case.

1 mark · Limited borrowing makes the venue's downside harder to absorb.

(d) **[10]**

Recommend an investment, taking account of both risk and the reliability of the figures.

10 marks · The stage gives CU 6,518 greater expected NPV at the stated probability and requires CU 80,000 less initial capital. The venue has a larger busy-case upside and may strengthen the brand through a fixed address. The probabilities rely on forecasts and the portable stage faces weather and site risks missing from the two-state model. Re-estimate demand, check venue financing and obtain site agreements. Select the stage if capital is tight or the quiet scenario is credible. A confidently higher probability of sustained busy trade can favour the venue. Use best fit. Award 0 for no creditable response, 1–2 for limited assertions, 3–4 for a partially explained point with weak application, 5–6 for developed analysis using case evidence, 7–8 for supported consideration of competing arguments and a reasoned decision, and 9–10 for sustained evaluation integrating specific evidence, a material limitation and a convincing judgement. Credit a well-supported alternative.

End of mark scheme · 50 marks